

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay.

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

A.

Cloud Auto-Scaling and Load Balancing

Cloud auto-scaling automatically increases or decreases the number of servers based on user demand. During seasonal sales, additional virtual machines are launched to handle the increased traffic, and when demand decreases, extra resources are removed to reduce costs.

Load balancing distributes incoming user requests evenly across multiple servers, preventing any single server from becoming overloaded. This ensures better performance and availability.

Reducing Downtime and Improving User Experience

- Prevents server overload during peak traffic.

- Minimizes application downtime.
- Provides faster response times.
- Ensures uninterrupted shopping experiences.

Role of Monitoring Tools

Monitoring tools continuously track CPU, memory, network traffic, and server health.

They trigger auto-scaling when predefined thresholds are exceeded and send alerts if any issues occur.

Real-World Example

During **Amazon's Great Indian Festival** or **Flipkart's Big Billion Days**, millions of customers access the websites simultaneously. Auto-scaling and load balancing allow the platforms to handle the increased demand without service interruptions.

Conclusion

Auto-scaling, load balancing, and monitoring ensure high availability, improved performance, and customer satisfaction during peak traffic.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

A.

Cloud Storage Durability and Redundancy

Cloud storage maintains multiple copies of data across different servers and data centres. If one storage device fails, the data remains available from another copy.

Encryption and Access Control

- Encrypt data during transmission (TLS/SSL).
- Encrypt stored data using AES-256.
- Implement Multi-Factor Authentication (MFA).
- Use Role-Based Access Control (RBAC) to restrict access.

Cost Advantages

- No investment in physical storage hardware.
- Pay only for the storage used.
- Lower maintenance and operational costs.
- Easy scalability without purchasing new equipment.

Disaster Recovery Example

If a company's primary data centre is damaged by a flood or fire, backup data stored in the cloud can be restored quickly, allowing business operations to continue with minimal downtime.

Conclusion

Cloud storage provides secure, durable, and cost-effective backup solutions while ensuring fast disaster recovery.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

A.

Role of Content Delivery Networks (CDNs)

A CDN stores copies of application content on multiple edge servers located around the world. Users receive data from the nearest server instead of the origin server.

Reducing Latency

- Shorter network distance.
- Faster content delivery.
- Reduced buffering.
- Lower network congestion.

Edge Servers

Edge servers cache frequently accessed content near users, improving application speed and reliability across different geographical regions.

Global Availability

Cloud providers maintain data centres in multiple regions, allowing applications to remain available even if one region experiences failures.

Example

Streaming platforms such as **Netflix** use CDNs to deliver videos quickly to users worldwide, while online gaming platforms use edge servers to reduce latency and improve gameplay.

Conclusion

CDNs and edge servers enhance application performance, reduce latency, and provide a better user experience globally.

4. A financial institution requires strict compliance and data security in cloud usage.

Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

A.

Regulatory Compliance and Auditing

Cloud providers support compliance with industry regulations by providing:

- Regular security audits.
- Compliance certifications.
- Detailed audit logs.
- Continuous monitoring.

Role of Logging, Monitoring, and Encryption

Logging

- Records user activities and system events.
- Helps detect suspicious behaviour.

Monitoring

- Tracks system health and security threats.
- Generates alerts for unusual activities.

Encryption

- Protects data during storage and transmission.
- Prevents unauthorized access.

Data Sovereignty

To meet data sovereignty requirements, organizations should:

- Store sensitive data in approved geographic regions.
- Follow country-specific legal and regulatory requirements.
- Restrict cross-border data transfers where necessary.

Example

A bank processing online fund transfers encrypts customer information, records every transaction in audit logs, and stores financial data within the country to comply with banking regulations.

Conclusion

Logging, monitoring, encryption, and compliance measures ensure secure financial

transactions while meeting legal and regulatory requirements.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

A.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Operating System	Share the host OS	Each VM has its own OS
Performance	Faster startup	Slower startup
Resource Usage	Low	High
Portability	High	Moderate
Deployment	Lightweight	Heavy

Role of Kubernetes

Kubernetes simplifies deployment by:

- Automating application deployment.
- Managing container scaling.
- Providing self-healing capabilities.
- Performing load balancing.
- Supporting rolling updates.

Portability

Containers package the application and its dependencies together, allowing the same application to run consistently across different cloud platforms without modification.

Microservices Example

An **online shopping application** can be divided into separate microservices such as:

- User Service
- Product Catalogue Service
- Order Service
- Payment Service
- Notification Service

Each microservice runs in its own container and can be independently deployed, updated, and scaled.

Conclusion

Containerization combined with Kubernetes provides faster deployment, efficient resource utilization, scalability, portability, and simplified application management across cloud environments.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity